

CredFlow

Digital Lending & Real-Time Credit Decisioning

An end-to-end digital lending platform designed to automate onboarding, KYC, credit assessment, risk-based decisioning and lender-side monitoring.

Real-time	decision engine
4	decision outcomes
5	test profiles
100%	decision auditability

THE CORE IDEA

Replace fragmented, manual underwriting with a transparent rule-driven lending journey — while giving higher-risk borrowers controlled alternatives instead of blanket rejection.

WHAT WAS BUILT

Component	Role in the platform
Borrower onboarding	Structured digital application and multi-step application flow
External-data layer	KYC, PAN, bureau, bank-statement and e-sign integrations designed around Indian financial infrastructure
Business Rule Engine	Real-time borrower risk evaluation and automated decisioning
Risk-based pricing	Reduced-principal or risk-premium alternatives for near-prime borrowers
Admin Command Center	Portfolio monitoring, application drill-down and decision audit trail

Project lens: CredFlow combines product design, financial mathematics, credit policy, automation and backend state management into one lending workflow.

A complete lending journey, designed around automated decisions.

The platform was architected to mirror a production-style Indian digital lending ecosystem, while the current demonstration uses simulated API calls.

Layer	Integration / capability	Purpose
Identity	Aadhaar eKYC / UIDAI / DigiLocker	Extract and verify Name, DOB, Gender and Address
Tax / identity	PAN / NSDL / Income Tax APIs	Identity verification and credit-record linkage
Credit bureau	TransUnion CIBIL / Experian	3-digit credit score and historical default data
Banking	Account Aggregator via vendors such as Perfios / Finbox	Salary credits, average balances and existing EMI obligations
Agreement	Aadhaar eSign / NSDL e-Gov	Legally binding digital agreement and live selfie capture

END-TO-END BORROWER FLOW

01	Onboard	Borrower enters personal, employment, income and loan details.
02	Verify	KYC, PAN, bureau and bank data are checked.
03	Calculate	EMI, total interest and DTI / FOIR are computed dynamically.
04	Decide	Business Rule Engine assigns KYC_FLAG, REJECTED, APPROVED_CONDITIONS or APPROVED.
05	Disburse	Approved applications proceed to final agreement and disbursement.
06	Monitor	Admin portal tracks applications, portfolio metrics and decision inputs.

DESIGN PRINCIPLE

The architecture separates data verification, financial calculation and credit decisioning so that every automated decision can be traced back to the inputs and rules that produced it.

Every lending decision starts with transparent financial math.

CredFlow dynamically calculates the borrower's repayment obligation and affordability before the Rule Engine evaluates risk.

Calculation	Formula	Purpose
EMI	$EMI = [P \times R \times (1+R)^N] / [(1+R)^N - 1]$	Calculates monthly repayment for principal, rate and tenure.
Total Interest	$Total\ Interest = (EMI \times N) - P$	Shows total cost of borrowing.
DTI / FOIR	$((Existing\ EMIs + New\ EMI) / Verified\ Monthly\ Income) \times 100$	Measures repayment burden.

WHAT THE ENGINE USES

P	Principal
R	Monthly rate
N	Tenure
DTI	Affordability

AFFORDABILITY GATE

Existing EMIs and verified income are sourced from mocked bank-statement data. If the proposed loan pushes DTI too high, the system flags the application rather than allowing a debt trap.

DECISION LOGIC AT A GLANCE

Condition	System response
Credit Score <600 OR DTI ≥50%	Immediate REJECTED status
Credit Score 650–749 OR DTI 40–49%	APPROVED_CONDITIONS with risk-based alternatives
Credit Score ≥750 AND DTI <40%	APPROVED at base rate
Aadhaar / PAN name mismatch	KYC_FLAG → Admin Portal for human verification

Four outcomes. No ambiguity.

The sequential Rule Engine removes manual underwriting from the standard path while retaining a human escalation route for identity exceptions.

Status	Trigger	Action
KYC_FLAG	Aadhaar name does not match PAN name	Pause auto-approval and route to Admin Portal
REJECTED	Credit Score <600 OR DTI ≥50%	Instant rejection to protect portfolio quality
APPROVED_CONDITIONS	Credit Score 650–749 OR DTI 40–49%	Offer controlled alternatives through risk-based pricing
APPROVED	Credit Score ≥750 AND DTI <40%	Frictionless approval for full requested amount at base rate

RISK-BASED PRICING — THE GRAY AREA

Option	Mechanism	Why it exists
DTI Optimisation	Reduce principal by 20%; keep base interest rate	Lower EMI and bring DTI back into a safer zone
Risk Premium	Approve full amount with +6% interest	Compensate lender for elevated near-prime risk

WHY THIS MATTERS

Instead of treating every non-prime borrower as a rejection, CredFlow converts measurable risk into a controlled offer — protecting the lender while preserving potential conversion.

Prime: remove friction. Near-prime: modify the economics. Sub-prime: protect the portfolio. KYC exception: escalate to a human.

Five borrower profiles were used to test the key edge cases.

The revised test suite covers prime, near-prime, sub-prime and returning-borrower situations, including an active top-up loan.

Case	Profile	Key calculation / signal	Engine outcome
01 • Robert	₹1.20L income; ₹5k existing EMI; CIBIL 782; ₹1.50L / 12 months	New EMI ₹13,327; total EMI ₹18,327; DTI 15.2%	APPROVED
02 • Chris	₹45k income; ₹8k existing EMI; CIBIL 715; ₹2.50L / 36 months	New EMI ₹9,041; total EMI ₹17,041; DTI 37.8%	APPROVED_CONDITIONS
03 • Benedict	CIBIL 580; 3 previous defaults	Sub-600 score triggers hard rejection	REJECTED
04 • Mark — Cleared Loan	Returning user; previous 6-month loan paid off; CIBIL 760; income ₹1.60L	Paid-off loan excluded from active DTI; verified identity bypasses standard KYC	APPROVED
05 • Mark — Active Top-Up	Active ₹3.00L loan; EMI ₹33,685; requests ₹1.50L top-up	New EMI ₹13,327; total EMI ₹47,012; DTI 29.38%	APPROVED

WHAT THE CASES DEMONSTRATE

Situation	System behaviour
Prime borrower	Frictionless approval when score and DTI are strong.
Near-prime borrower	Controlled alternatives rather than flat rejection.
Sub-prime borrower	Hard rejection protects portfolio quality.
Returning borrower	Historical paid-off loan does not inflate active DTI.
Top-up borrower	Old and new EMIs are aggregated correctly while remaining separately itemised in the schedule ledger.

EDGE-CASE COVERAGE

The revised profiles test the core decision paths and the most mathematically complex returning-borrower scenario.

From borrower workflow to lender command center.

The backend monitoring layer ties together application state, real-time synchronisation and transparent credit-decision auditing.

Capability	Implementation / purpose
Frontend State Machine	React Context API manages the complex multi-step application state.
Real-Time Cross-Tab Syncing	Cross-tab local-storage events sync borrower milestones, such as agreement signing, to the Admin Portal in real time.
Precision Tracking	Custom tracking logic prevents ghost applications when users simply view their dashboard.
Admin Command Center	Credit officers view Approval Ratios, Total Disbursed and individual applications.
Decision Audit	Admins can audit the exact DTI math, CIBIL score and Bank Statement analysis used by the Rule Engine.

RETURNING-BORROWER ACCOUNTING

₹33,685	existing EMI
₹13,327	top-up EMI
₹47,012	combined EMI
29.38%	DTI

TRANSPARENCY BY DESIGN

The borrower experience is automated, but the lender retains visibility into exactly why a decision was made — supporting auditability and compliance.

WHY THE ARCHITECTURE MATTERS

CredFlow is not only an application form. It links data capture → verification → financial calculation → risk rules → offer generation → application tracking → lender audit into a single decision flow.

More than a lending UI. A decision system.

CredFlow brings together the customer journey, financial mathematics, risk policy and lender monitoring into one coherent digital lending architecture.

Challenge	CredFlow response
Slow, manual underwriting	Real-time Business Rule Engine
Fragmented verification	Integrated KYC / PAN / bureau / banking design
Opaque affordability	Dynamic EMI, interest and DTI / FOIR calculations
Binary approval mindset	Risk-based pricing for near-prime borrowers
Weak audit trail	Admin drill-down into every decision input
Complex returning borrowers	Historical-loan and top-up-aware DTI calculations

4	decision outcomes
5	test profiles
2	returning-loan states
1	integrated rule engine

KEY TAKEAWAY

CredFlow demonstrates how lending decisions can be made faster without making them opaque: automate routine approvals, control risk through explicit rules, offer structured alternatives to near-prime borrowers, and escalate identity exceptions to humans.

Portfolio-ready summary: Developed an end-to-end digital lending platform featuring real-time KYC, credit and affordability checks, a business rule engine, risk-based pricing, returning-borrower logic and an admin command center for transparent decision monitoring.

Source: Updated Project CredFlow — Final Project Report.